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PAPER_243 — NGC 3603 Full MUGE: Time-Varying Mass $M(t)$ and Additive Cavity Pressure $P(t)/\rho$

Date: October 2025 ## NGC 3603 Extreme Young Star Cluster — Complete 10-Term Master Universal Gravity Equation

Author: Daniel T. Murphy

Framework: Unified Quantum Field Framework (UQFF) v4.10

Calculator Class: NGC3603FullMUGECavityPressureCalculator (CP3, Session 60)

Encoded By: Grok (xAI), October 2025 (C++ source); Python CP3 integration March 2026

Version: 1.0 | **Session:** 60 | **PAPER Number:** 243

<!-- UQFF constants: $\kappa = 5.0\text{e-}4 \text{ day-1}$, $[\text{SSq}] = 0.57$, $M_{\text{UQFF}} = 1.43\text{e1 TeV}$ -->

Abstract

This paper presents the complete 10-term Master Universal Gravity Equation (MUGE) for the NGC 3603 extreme young star cluster, incorporating two novel mathematical elements not previously captured in CP3:

- Time-varying mass** $M(t) = M_0(1 + \dot{M}_{\text{factor}} e^{-t/\tau_{\text{SF}}})$ — exponential star-formation inflow driving cluster mass growth over the first few Myr.
- Additive cavity pressure acceleration** $P(t)/\rho_{\text{fluid}}$ where $P(t) = P_0 e^{-t/\tau_{\text{exp}}}$ — the ionized cavity pressure from O/B-star UV/wind feedback expressed as an acceleration term alongside (not as a modifier of) the gravitational field.

These are **expressly distinct** from CP3 class 88 (NGC3603StellarPressureModulationCalculator, Session 55), which uses pressure as a multiplicative suppressor $g \propto (1 - P(t))$. The present paper treats pressure as an independent additive acceleration channel.

2. System Parameters

Parameter	Symbol	Default Value	Units
Initial mass	M_0	$4 \times 10^5 M_{\odot}$	kg
Cluster radius	r	9.5 ly	m
Star-formation timescale	τ_{SF}	1 Myr	s
Initial cavity pressure	P_0	4×10^{-8}	Pa
Pressure expansion timescale	τ_{exp}	1 Myr	s
Wind density	ρ_{wind}	10^{-20}	kg/m3
Wind velocity	v_{wind}	2×10^6	m/s

Parameter	Symbol	Default Value	Units
Fluid density	ρ_{fluid}	10^{-20}	kg/m ³
Magnetic field	B	10^{-5}	T

3. Novel Mathematical Elements

3.1 Time-Varying Cluster Mass

$$M(t) = M_0 \left(1 + \dot{M}_{\text{factor}} e^{-t/\tau_{SF}} \right)$$

where $\dot{M}_{\text{factor}}$ is the dimensionless peak accretion enhancement. At $t = 0$, $M(0) = M_0(1 + \dot{M}_{\text{factor}})$ (maximum infall); at $t \gg \tau_{SF}$, $M \rightarrow M_0$ (steady state). The **star-formation rate** driving this is:

$$\frac{dM}{dt} = - \frac{M_0 \dot{M}_{\text{factor}}}{\tau_{SF}} e^{-t/\tau_{SF}}$$

The **star-formation efficiency** at time t is:

$$\varepsilon_{SF}(t) = \frac{M(t) - M_0}{M_0} = \dot{M}_{\text{factor}} e^{-t/\tau_{SF}}$$

3.2 Additive Cavity Pressure Acceleration

The O/B-star population inflates an ionized HII cavity. The cavity pressure decays exponentially as the molecular cloud is dispersed:

$$P(t) = P_0 e^{-t/\tau_{\text{exp}}}$$

This converts to a specific acceleration (per unit mass of ambient gas):

$$T_{\text{pressure}} = \frac{P(t)}{\rho_{\text{fluid}}} = \frac{P_0 e^{-t/\tau_{\text{exp}}}}{\rho_{\text{fluid}}}$$

Dispersal condition: The cluster disperses its natal cloud when the cavity-pressure acceleration exceeds local gravity:

$$\frac{P(t_{\text{disp}})}{\rho_{\text{fluid}}} = T_1 \Rightarrow t_{\text{disp}} = \tau_{\text{exp}} \ln! \left(\frac{P_0}{\rho_{\text{fluid}} \cdot T_1} \right)$$

4. Full 10-Term MUGE

$$g_{\text{NGC3603}}(r, t) = T_1 + T_2 + T_3 + T_4 + T_q + T_{\text{fl}} + T_{\text{osc}} + T_{\text{DM}} + T_{\text{wind}} + T_{\text{pressure}}$$

Term 1 — Base with M(t), Hubble, magnetic:

$$T_1 = \frac{GM(t)}{r^2} (1 + H_0 t) (1 - B/B_{\text{crit}})$$

Note: Unlike class 88 which applies $(1 - P)$ here, the full MUGE leaves T_1 unmodified and adds P explicitly as T_{10} .

Term 2 — UQFF unified field:

$$T_2 = (U_{g1}(t) + U_{g4}(t))(1 + f_{TRZ}), \quad U_{gk}(t) \propto M(t)/r^2$$

Term 3 — Cosmological constant:

$$T_3 = \frac{\Lambda c^2}{3}$$

Term 4 — EM with vacuum ratio:

$$T_4 = \frac{qv_{\text{gas}}B}{m_p} \left(1 + \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} \right) s_{\text{EM}}$$

Term 5 — Quantum uncertainty:

$$T_q = \frac{\hbar}{\sqrt{\Delta x \Delta p}} \psi, \frac{2\pi}{t_H}$$

Term 6 — Fluid acceleration:

$$T_{\text{fl}} = \frac{\rho_{\text{fluid}} V g_{\text{base}}(t)}{M(t)}$$

Term 7 — Two-mode oscillation:

$$T_{\text{osc}} = 2A \cos(kx) \cos(\omega t) + \frac{2\pi}{t_{\text{Gyr}}} A \cos(kx - \omega t)$$

Term 8 — Dark matter with $3GM(t)/r^3$ tidal perturbation:

$$T_{\text{DM}} = \frac{(M(t) + M_{\text{DM}})(\delta\rho/\rho + 3GM(t)/r^3)}{M(t)}$$

Term 9 — Stellar wind ram pressure:

$$T_{\text{wind}} = \frac{\rho_{\text{wind}} v_{\text{wind}}^2}{\rho_{\text{fluid}}}$$

Term 10 — Cavity pressure (additive, novel):

$$T_{\text{pressure}} = \frac{P_0 e^{-t/\tau_{\text{exp}}}}{\rho_{\text{fluid}}}$$

5. Distinction from Prior Art

Feature	Class 88 (Session 55)	This class (Session 60)
Mass	Fixed M	$M(t) = M_0(1 + \dot{M}e^{-t/\tau})$
Pressure form	Multiplicative $(1 - P(t))$ modifier	Additive $P(t)/\rho_{\text{fluid}}$ term
Physics	UV/wind pressure suppresses gravity	Cavity pressure drives independent acc.
Terms	4 (base, U_g , Λ , wind)	10 (all MUGE terms)
Quantum term	Not present	T_q present
Fluid term	Not present	T_{fl} present
2-mode oscillation	Not present	Standing + traveling wave
DM pert2	Not present	$3GM(t)/r^3$

Feature	Class 88 (Session 55)	This class (Session 60)
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6. Cross-Validation: Class 88 vs Class 111 Regime

For small P_0 and short $t \ll \tau_{\text{exp}}$:

$$P(t) \approx P_0 \Rightarrow T_{\text{pressure}} \approx P_0 / \rho_{\text{fl}}$$

For Class 88, the $(1 - P)$ effect on T_1 :

$$\Delta T_1 = -T_1^{\text{Class88}} \cdot P \approx - \underbrace{\frac{GM}{r^2}}_{\mu_s \nabla(M_s/r)} P$$

The two are **equivalent only in the limit** where $P_0 / \rho_{\text{fl}} = (\mu_s \nabla(M_s/r))P$, i.e., when $P_0 = GM\rho_{\text{fl}}/r^2$. At default values of NGC 3603, this equality does not hold — the additive and multiplicative forms predict **different total** g and different dispersal timescales.

7. Numerical Example

At $t = 0.5$ Myr, default parameters:

$$M_{\text{dot}}(t) = 1.0 \times e^{-0.5} \approx 0.607 \Rightarrow M(t) \approx 1.607 M_0$$

$$P(t) = 4 \times 10^{-8} \times e^{-0.5} \approx 2.43 \times 10^{-8} \text{ Pa}$$

$$T_{\text{pressure}} = \frac{2.43 \times 10^{-8}}{10^{-20}} = 2.43 \times 10^{12} \text{ m/s}^2$$

The cavity pressure acceleration dominates all other terms at early times — consistent with the observational picture of NGC 3603's OB-star cluster rapidly dispersing its natal cloud within ~ 3 Myr (Harayama et al. 2008; Pang et al. 2013).

8. Simulation Recommendations

1. τ_{SF} **sweep** (0.5–5 Myr): Track $M(t)$ peak mass vs. cluster binding energy
 2. P_0 **sweep** (10^{-9} – 10^{-6} Pa): Determine cavity pressure dispersal threshold
 3. **Term10 vs Term9 cross-over**: Find t^* where $T_{\text{pressure}}(t^*) = T_{\text{wind}}$
 4. M_{dot} **sweep** (0.1–2.0): Map star-formation efficiency vs. gravitational retention
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9. Integration Into UQFF Pipeline

- **CP3 class:** NGC3603FullMUGECavityPressureCalculator (113th AST node, 112th calculator, Session 60)
 - **Aggregator:** v2.4.0 — registered in CP3_CALCULATORS
 - **Source:** Doc 11 C++ class NGC3603 (Grok/xAI, October 2025)
 - **Complementary class:**
 - Class 88 NGC3603StellarPressureModulationCalculator — multiplicative pressure form (Session 55)
 - Class 110 RingsOfRelativityEinsteinLensingMUGECalculator — companion new class (PAPER_242)
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10. References

- Harayama Y. et al. (2008): ApJ 675, 1319 — NGC 3603 stellar mass function
 - Pang X. et al. (2013): ApJ 764, 73 — star formation history
 - Portegies Zwart S. et al. (2010): Nature 464, 203 — young massive clusters
 - Murphy, D.T. (2025): UQFF Manuscript — Doc 11 (NGC 3603 MUGE, October 2025)
 - Grok (xAI): C++ class NGC3603, encoded October 2025
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PAPER_243 / Session 60 / CP3 class 112 (NGC3603FullMUGECavityPressureCalculator) / UQFF v4.10

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_{Bi}_i\}}$ jet modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SC_m vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SC_m vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile), PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044 (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c} \right)^2 \right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad (\text{vs standard } b = 0.20)$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\text{--}4\%$ at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^{(3)} \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$ sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$ (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)

Sector	Domain	Late-Corpus Result
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	$F_{\{U_Bi_i\}}$ buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **galaxy-rotation** sector of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{rot}})(\partial^\mu \phi_{\text{rot}}) - V(\phi_{\text{rot}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{rot}}) = \frac{1}{2}m^2\phi_{\text{rot}}^2 + \frac{\lambda}{4!}\phi_{\text{rot}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{rot}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{rot}}} = v_c^2/r - \mu_s \nabla(M_s/r) - F_{U_Bi_i}/(m \cdot r) + \rho_{\text{vac},[\text{SCm}]} \cdot \Omega^2 r = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{rot}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(- \exp! \left(- \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.108 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This

threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 7, \quad n_{\text{channel}} = 10/26$$

Since $p_{\text{DVP}} = 7$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **109 yr** (disk settling timescale):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos! \left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.108	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 7$	PASS Sub-threshold
BSH layers	26 harmonic terms	$j = 1 \dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ $2(UQFF_{vacuumterm}) 1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	PASS Consistent	
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature	$F_{U_Bi_i}$ unique gravitational correction	Not yet measured	Future gravita- tional wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{U_Bi_i}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1039	SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model
PAPER_1040	SCm Cluster Merger Shock Mach Number Phonon Damping
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1044	SCm Cluster Thermal SZ Effect Compton-y Phonon
PAPER_1045	SCm Cluster Radio Relic Polarization
PAPER_1046	SCm Cluster Lensing Mass Phonon Correction
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1050	MUGE $F_{U_Bi_i}$ Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

11 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_{kozima\_ramanujan\_appendices}.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_{s26_coupling}.py</code>	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_{scm_cross_section}.py</code>	S_{26} -modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor ($1 + [SSq]^n/26$)
<code>kozima_{wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{SCm} = N_n \times \sigma_n^{SCm}(\omega) \times \Phi_{\text{phonon}} \times (F_{\{U,Bi\}}/F_U - 1)$ where $\sigma_n^{SCm}(\omega, n) = \sigma_0 \times \exp[-(\omega - \omega_{SCm})^{2/(2 \times \Gamma_2)}] \times (1 + [SSq]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_{polylog_s26}.py</code>	$Li_{26}([SSq])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_{wstp_kernel}.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1 - 2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} \times Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
<code>mock_{theta_q26}.py</code>	$f_{26}(q)$, $\phi_{26}(q)$, $\psi_{26}(q)$ q-series	Proper q-Pochhammer $(a; q)_n$

Core equations: $-f_{26}(q) = \sum_{n=0}^{25} q^{\{n2\}} / (-q; q)_n$ - $\phi_{26}(q) = \sum_{n=0}^{25} q^{\{n2\}} / (-q^2; q^2)_n$ - $\psi_{26}(q) = \sum_{n=1}^{26} q^{\{n2\}} / (q; q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
<code>ramanujan_{pi_uqff}.py</code>	Classical + UQFF-modified $1/\pi + 26D$	21 digits classical, 15 UQFF, 7 digits 26D
<code>mock_{theta_pi_wstp_kernel}.py</code>	11-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n \times (1103 + 26390n) \times W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-\kappa_{\text{pai}} \times n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate

Symbol	Value	Description
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\_CoAnQi}.cpp`, and Wolfram kernels (`uqff_{kozima\_kernel}.wl`, `uqff_{s26\_kernel}.wl`, `uqff_{mock\_theta\_pi\_kernel}.wl`).

References

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
4. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x
5. McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei*. ARA&A **45**, 117 — arXiv:0709.4098 — doi:10.1146/annurev.astro.45.051806.110625
6. Murphy, D. (2026). *Master Universal Gravity Equation (MUGE): DPM-Driven Gravity Framework*. Star-Magic Whitepaper Series — github.com/Daniel8Murphy0007/Star-Magic